

Alex Mayhew

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Available for part-time, contract, or full-time remote work.

Technology Specialist: Web, SEO & Automation

Technology generalist who spent nine years as the one-person tech department for a Martha's Vineyard solar company: designing ~200 residential PV systems, handling permitting through historic and conservation boards, building and ranking the company website, coding its custom lead-intake system, and running office IT. Now self-employed doing web, automation, and AI-assisted work for clients and my own products. Used to being the only technical person in the room, and to explaining technology to people who aren't technical.

EXPERIENCE

Harvest Sun Solar, Martha's Vineyard & Cape Cod, MA

Solar Designer & Technology Specialist · March 2016 – December 2025 (company closed)

- Designed ~200 residential PV systems in Aurora Solar (module selection, stringing and component specification, fire-setback compliance, and production modeling), building custom 3D layouts from 2D blueprints, plus SketchUp models presented to historic-district and conservation-commission boards.
- Handled permitting end to end across Martha's Vineyard and Cape Cod (standard, historic-district, and conservation-commission review), and presented projects in person to town boards and building departments.
- Contributed design work on commercial projects of 2–3 MW, including two designed through proposal and permitting that are slated for construction.
- Built the company's lead-capture and customer-onboarding system solo (React/TypeScript, Node/Express, PostgreSQL): multi-step intake with address autocomplete and satellite-map property confirmation (Google Maps APIs), service-area validation, electric-bill collection, automated customer emails (SendGrid), and two-way sync into the SolarNexus CRM. Containerized and deployed with Docker/Traefik on self-managed infrastructure.
- Built and ran the company WordPress site, reaching #1 Google rankings for the company's core Martha's Vineyard solar searches, and managed the full local-marketing presence (Google Business Profile, EnergySage, Angi/HomeAdvisor, Yelp, BBB), then handed the site to an outside agency and supported them on technical implementation.
- Ran IT for the 5-person office and field crews: Microsoft 365 administration, workstation setup and repair, network and ISP troubleshooting, printer and peripheral support, vendor coordination, and first-line support; administered the SolarNexus CRM and ran a six-month GoHighLevel evaluation.
- Handled post-install troubleshooting directly with homeowners: remote diagnosis and coordination with equipment vendors.

Self-Employed: Web, Automation & AI Services (Mayhew Technology LLC), Remote

January 2026 – present

- Sole developer on a maritime data-intelligence platform for a private investment firm (engagement under NDA). Took over a demo-stage build and grew it into a production platform (Next.js + FastAPI + PostgreSQL/TimescaleDB, 100+ maritime data sources), split the codebase into two independently deployed products under a hard deadline, and closed out with a written engineering handoff plus a staffing recommendation for the client's incoming team.
- Provision, harden, and operate production Linux servers for my own products and client deployments, documented as reproducible install-and-verify runbooks: nginx, systemd, fail2ban/AppArmor/auditd, automated off-site backups (restic to Cloudflare R2), and Prometheus/Grafana monitoring.

- Built web apps for local small businesses: an offline-first time-tracking and fleet app for a trucking and farm-services company, running in production on a server I manage, and an active e-commerce site for a bakery (Next.js, Stripe, CMS).
- Built data-extraction tools for freelance clients, including a tax-auction property scraper for a real-estate investor that cross-references a 2,400-property county auction book (PDF) against a 1.5M-record ArcGIS tax-roll API into investment-ready spreadsheets, plus a scraping platform with headless-browser automation, proxy rotation, per-domain rate limiting, and queued job processing (Python, TypeScript, Puppeteer, Redis).
- Build and support my own products (WordPress, Next.js, Python), including an AI voice/audio platform and a commercial SaaS starter kit, plus automations and AI-assisted workflows with n8n (Zapier/Make-equivalent), REST APIs, and LLM tools.
- Publish a technical blog and newsletter (70+ articles) on websites, performance, and SEO.

SKILLS

Web: WordPress, HTML/CSS/JavaScript, React/TypeScript, Next.js, hosting/DNS/domain and email administration, Docker

SEO & Local Marketing: technical and local SEO, Google Search Console, GA4, keyword research, schema/structured data, Google Business Profile, listings management (EnergySage, Angi, Yelp, BBB), answer-engine optimization (AEO/GEO)

Automation & Scripting: n8n (Zapier/Make-equivalent), Python scripting, web scraping and data extraction (Puppeteer/Playwright, proxy rotation, rate limiting), REST APIs, webhooks, JSON, LLM tools (Claude, ChatGPT), AI-assisted development (Claude Code), self-hosted LLMs (Ollama), vector/graph databases (Qdrant, Memgraph)

Support & IT: end-user and remote support, hardware and software troubleshooting, device setup, Windows/macOS/Linux, Microsoft 365 administration, CRM administration (SolarNexus, GoHighLevel), technical documentation and runbooks, customer service

Servers & DevOps: Linux server administration (Ubuntu/Debian), nginx, systemd, Docker/Traefik, security hardening (fail2ban, AppArmor, auditd), CI/CD (GitHub Actions, GitLab CI), backups (restic, Cloudflare R2), monitoring (Prometheus, Grafana)

Solar: Aurora Solar, Energy Toolbase, SketchUp, residential PV design, stringing and component specification, fire setbacks, permitting and AHJ processes, historic-district and conservation approvals

EDUCATION

Self-taught in programming, web design, and IT since age 14, and still how I pick up new tools. High school diploma.